The artery size hypothesis: a macrovascular link between erectile dysfunction and coronary artery disease.

Erectile dysfunction (ED) is defined as the inability to achieve or maintain an erection satisfactory for sexual performance. Evidence is accumulating to consider ED as a vascular disorder. Common risk factors for atherosclerosis are frequently found in association with ED, and ED is frequently reported in vascular syndromes, such as coronary artery disease (CAD), hypertension, cerebrovascular disease, peripheral arterial disease, and diabetes mellitus. Finally, similar early impairment of endothelium-dependent vasodilatation and late obstructive vascular changes has been reported in both ED and other vascular syndromes. Recently, we proposed a pathophysiologic mechanism to explain the link between ED and CAD called the artery size hypothesis. Given the systemic nature of atherosclerosis, all major vascular beds should be affected to the same extent. However, symptoms rarely become evident at the same time. This difference in rate of occurrence of different symptoms is proposed to be caused by the different size of the arteries supplying different vascular beds that allow a larger vessel to better tolerate the same amount of plaque compared with a smaller one. According to this hypothesis, because penile arteries are smaller in diameter than coronary arteries, patients with ED will seldom have concomitant symptoms of CAD, whereas patients with CAD will frequently complain of ED. Available clinical evidence appears to support this hypothesis.